

Answer Submitted.

X


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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Theory of Computation (course)

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week - 0 ()

● Practice:
Week 0 :
Assignment 0
(assessment?
name=146)

Week - 1 ()

Week - 2 ()

Week - 3 ()

Week - 4 ()

Week 0 : Assignment 0

Your last recorded submission was on 2025-07-24, 11:48 IST

1) The number of bit strings of length 8 that either start with 1 or end with 00 is?

1 point

- ☐ 32
☐ 128
☐ 160
☒ 192

No, the answer is incorrect.

Score: 0

Accepted Answers:

160

2) Let G be a connected planar graph with 10 vertices. If the number of edges on each face is 3, then the number of edges in G is?

1 point

- ☐ 20
☐ 24
☐ 32
☒ 36

No, the answer is incorrect.

Score: 0

Accepted Answers:

24

3) Which of the following is correct?

1 point

- ☒ The number of odd degree vertices in an undirected graph is odd.
☐ The number of odd degree vertices in an undirected graph is even.
☐ The number of even degree vertices in an undirected graph is even.



Week - 5 ()

Week - 6 ()

Week - 7 ()

☐ The number of even degree vertices in an undirected graph is odd.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The number of odd degree vertices in an undirected graph is even.

4) Consider the binary relation $R = \{(x, x), (y, y), (x, z), (z, x)\}$ on the set $\{x, y, z\}$. Which one of the following is TRUE?

1 point



R is reflexive



R is symmetric



R is anti-symmetric



R is anti-reflexive

No, the answer is incorrect.

Score: 0

Accepted Answers:

R is symmetric

5) How many onto (surjective) functions are there from an n -element set ($n \geq 2$) to a 2-element set? **1 point**



$2^n - 2$



$2^n - 1$



2^n



$2(2^n - 1)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2^n - 2$

6) What is the total number of possible symmetric relations on a set of 3 elements? **1 point**



2^9



2^6



2^3



2

No, the answer is incorrect.

Score: 0

Accepted Answers:

2^6

7) How many even numbers of 2 digits are there which have both digits distinct? **1 point**



41

☐ 40☐ 42☒ 43

No, the answer is incorrect.

Score: 0

Accepted Answers:

41

8) How many 4-digit numbers can be formed from the digits $\{1, 2, 3, 4, 5, 6, 7\}$ which **1 point** are divisible by 5 (none of the digits are repeated)?

☒ 24☐ 36☐ 120☐ 720

No, the answer is incorrect.

Score: 0

Accepted Answers:

120

Check Answers and Submit

Your score is: 0/8

